

Conventional force-aware VLA

Vision / language / state / force

Full-action prediction

task motion + contact adjustment

Contact changes can outpace
action updates

failure frame

Reference error

failure frame

Stale reference

failure frame

Jamming

failure frame

Nonlocal recovery

ForceDelta-VLA

Teacher-defined force residual

Frozen teacher · matched context/noise

Force-conditioned
prediction

Learned missing-force
prediction

Force-responsive
residual target

Fast residual execution

Force / state

Fast residual policy

Cached reference

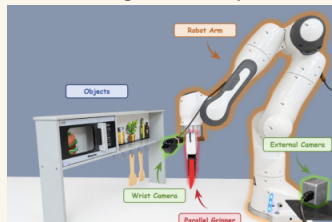
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Commanded action

Distill

Robot platforms

Single-arm setup



Bimanual setup

